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3 October 1986
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8 November 1985
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INT. AMENDMENT 1(NAVY-SH)
3 December 1984

MILITARY SPECIFICATION

CABLES AND CORDS, ELECTRICAL, LOW SMOKE, FOR SHIPBOARD USE
GENERAL SPECIFICATION FOR

This specification is approved for use by all Departments and Agencies of the Department of Defense.

PAGES 1 through 4

* Remove old pages 1 through 4 and replace with new pages 1 through 4.

PAGE 7

After ASTM D 470 - Standard Methods ..... Cable: Add "D 1248 - Standard Specification for Polyethylene Plastics Molding and Extrusion Materials."

PAGE 9

* 3.3.5, first sentence, third line: Delete "may" and substitute "shall".

The attached insertable replacement pages listed below are replacements for stipulated pages. When the new pages have been entered in the document, insert the amendment as the cover sheet to the specification.

Replacement page	Page replaced
1	1
2	2
3	3
4	4

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* Table IV, Note 1/, after the last sentence: Add

"Strand construction other than specified above shall be approved by NAVSEA."

TABLE VIII. Navy standard sizes. Under "Bunch-lay-stranded:" Add: "9(90) 90 .010 .120 9., 045 8,864 1.22 1.28 28".

Table XI: Delete the Background or base colors for conductor or group number 3, 4, 5 and 6 and substitute:

"Red
Green
Orange
Blue"

Table XIII: Delete and substitute:

"TABLE XIII. Qualification samples and groups.

Qualification test				
Group number	Additional cable types and sizes for flame propagation test only	Additional cable types and sizes for those tests only which are applicable to overall shielded cables	All test specified in the applicable specification sheet type and size	Types comprising the group 1/
1	None	None	LSTHOF-9	Navy Standard sizes 42 and smaller of types LSDHOF, LSTHOR and LSFHOF except type-and-size LSTHOF-14.
* 2	None	None	LSTHOF-150	Navy Standard sizes 60 and larger of types LSDHOR, LSTHOF, and LSFHOF type LSCVSF; type LSTHOF-14.
3	LSTPNW-20	LSMUS-14 or LSMS-37	LSMDU-23	LSMDU, LSMDY, LSMRI, LSPBTMU, LSMU, LSMS, LS2SJ, LS3SJ, LS4SJ and LSTPNW.

See footnote at end of table.

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"TABLE XIII. Qualification samples and groups. - Continued

Qualification test				
Group number	Additional cable types and sizes for flame propagation test only	Additional cable types and sizes for those tests only which are applicable to overall shielded cables	All test specified in the applicable specification sheet type and size	Types comprising the group 1/
4	None	None	LSMMOP-5 or LSTTOP-10 or LSMHOF-30	LSMMOP, LSTTOP, LSMHOF, LSDCOP and LSTCOP.
5	None	None	LSMCOS-6	LSMCOS
6	LSTSGU-14	LSMSCS-10	LSMSCU-10	LSMSCU, LS7SGU, LSTCJU, LSTCTU and LSTTSU. AWG sizes 7 and smaller of LSDSGU, LSTSGU and LSFSGU.
7	LSTSGU-50	None	LSTSGU-150	AWG sizes 3 and larger of LSTSGU, LSSSGU, LSDSGU, LSFSGU, LS6SGU and LS5KVTSGU.
8	LSPI-12	None	LSTCTX-7	LSTCTX, LSTCJX, LSTCKX, LSPI, LSDPS, LSTPS, LSFPS and LS7PS.
9	None	None	LSTTRS-8	LSTTRS
10	LSTNW-14 and LSTNW-50	None	LSTNW-150	LSTNW, LSDNW, LSFNW, LSMNW, LS8NW6 and LS4NW8.
11	None	LS1S50MUS-16 or LS2AUS-40	LS1S50MU-16 or LS2AU-40	LS1S50MU, LS2AU, LS3SU, LS3U, LS1S75MU, LS1SMU, LS1SAU, LS1SU, LS3SF, LS2U and LS2CS.
12	None	LS3SWUS-10 or LS2UWS-42	LS2SWAU-10	LS2SWAU, LS1SWU, LS2SWU LS3SWU, LSECM, LS2WAU, LS1SMWU, LSSRW, LSDRW, LSTRW, LS2SWL and LS2U

1/ Includes all sizes and variations of the cable type."

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PAGE 31

* TABLE XIV. Insert the following at the bottom of Group A:

Crack resistance	specification sheet	4.7.33

PAGE 56

* 4.7.15.2 (a), second line: Delete ..."and 10"...

* 4.7.15.2 (b), delete and substitute:

"(b) A ribbon burner, which accepts gas and air from the gas and air supply via a single, 1/2-inch nominal pipe set vertically into its underside, which thoroughly blends the gas and air within an internal mixing chamber, which allows the blended gas and air mixture to exit for burning via a multiplicity of holes in its top, and which shall be fabricated in accordance with figures 8 and 9. The mixing chamber, shall be fabricated from number 16 United States Standard Gage (USSG) steel, shall be horizontally centered within the burner, and shall be airtight except for the specified holes on its two sides and through the inside of the 1/2-inch nominal pipe. The burner exterior shall be fabricated from number 14 USSG steel, and shall be airtight except for the specified holes on its top and the specified holes in the mixing chamber."

PAGE 70

4.7.27, item (b): Add "with a nominal wall thickness of 0.020 inch of insulation material."

PAGE 71

4.7.29, second sentence: Delete and substitute: "For insulation materials, the toxicity index shall be calculated on the mass of insulation on 1-meter length of finished wire size 22 AWG with a nominal wall thickness of 0.020 inch of insulation material."

* 4.7.31, second sentence: Delete and substitute: "There shall be 1-inch bench marks, 1-inch jaw separation and a rate of jaw travel of 10 inches per minute. For cross-linked polyethylene insulation meeting the criteria of ASTM D 1248 for type II, III and IV insulation the rate of jaw travel shall be 2 inches per minute."

Add as new paragraphs:

* "4.7.33 Crack resistance. This test shall determine the ability of glass braid covering to withstand bending, and to determine the propensity of glass braids to puncture overlying covering.

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* "4.7.33.1 Specimen. Each specimen shall consist of a single insulated conductor (including insulation covering) of a length not less than that specified in table XXII. Specimens shall be removed from completed cable or cord as follows:

(a) Cables or cords containing four or fewer conductors: one specimen shall be removed from each insulated conductor.
(b) Cables or cords containing more than four but fewer than 21 conductors: one specimen shall be removed from each of four different insulated conductors (a total of four specimens).
(c) Cables or cords containing 21 or more conductors: the number of specimens shall be equal to the square root (rounded, if necessary, to the nearer whole number) of the total number of conductors. Each specimen shall be removed from a different insulated conductor.

"An additional specimen, which shall consist of a piece of insulated conductor (including insulation covering) of not less than 10 feet in length, may be required (see 4.7.33.5).

* "TABLE XXII. Specimen and mandrel dimensions for the crack resistance test.

Conductor size (AWG)	Minimum specimen length, (inches)	Mandrel diameter (inches)
7	24	1-1/2 ± 1/16
9	18	1 ± 1/16
10	12	1/2 ± 1/32
14	12	1/2 ± 1/32
16	8	1/3 ± 1/64
18	8	1/3 ± 1/64
Greater than 18	6	1/4 ± 1/64

* "4.7.33.2 Special apparatus. Apparatus shall include the following:

(a) Mandrel shall be a rigid, smooth cylinder, of the diameter specified in table XXII, and the length and construction for the specimen wrapping as specified in 4.7.33.3.
(b) An aqueous solution of gentian violet, of a quantity and concentration for the staining as specified in 4.7.33.3 or 4.7.33.5, as applicable.

* "4.7.33.3 Procedure. Each specimen shall be tightly hand-wrapped, for not fewer than three circumferential and continguous, 360-degree turns around the mandrel. Following this, each specimen shall be removed from the mandrel and continguous, 360-degree turns in the opposite lateral direction. Each specimen shall next be removed from the mandrel and straightened, to the extent that the specimen is nowhere bent along a radius of less than 12 times the maximum measured specimen diameter. Each straightened specimen shall then be submerged, except for its extreme ends (which shall remain exposed to the air), in the gentian violet for a period of not less than 30 minutes, and shall then be removed from the gentian violet and visually inspected for defects.

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* "4.7.33.4 Observation. Specimen failure shall be construed if the insulation covering of any specimen exhibits visible cracking or peeling which allows penetration of the gentian violet. If the insulation covering of any specimen exhibits an average of more than one pinhole, which allows penetration of the gentian violet for each 12 inches of length, then the further procedure as specified in 4.7.33.5 shall be performed.

* "4.7.33.5 Further procedure (may not be required, see 4.7.33.4). An additional specimen, which shall consist of a piece of insulated conductor (including insulation covering) of not less than 10 feet in length, shall be prepared. This entire specimen, except for its extreme ends (which shall remain exposed to the air) shall be submerged in gentian violet for a period of not less than 30 minutes. To facilitate submersion, the specimen may be coiled to the extent that is a nowhere bent along a radius of less than 12 times the maximum measured specimen diameter. The specimen shall then be removed from the gentian violet and visually inspected for defects.

* "4.7.33.6 Further observation. Specimen failure shall be construed if the insulation covering of the specimen exhibits an average of more than one pinhole, which allows penetration of the gentian violet for each 12 inches of length."

PAGE 80

* 4.8.5.4(b), after last sentence and before (additional measurements ... see (d).) add: "If the specimen does not incorporate an overall shield or armor over the jacket the test shall be made after placing the reel of cable in water for a period of not less than 1 hour."

PAGE 81

* 4.8.5.4(d), delete the first sixteen lines (through "... in the following formula:") and substitute:

"(d) Specimens incorporating double overall shields: one shield shall be electrically connected to one terminal of the megohmmeter and the other shield shall be connected to the other terminal. The resistance measurement shall be performed as specified. (Additional measurements required, see (a)).

(e) Specimens incorporating single or multiple insulated conductors without shields or armor and other specimens requiring measurements to water (see 3.1): the specimens shall be submerged for not less than 1 hour, except for its extreme ends (which shall remain exposed to the air), in the water tank. The single conductor, or on multiconductor cables, the conductors which are to one terminal of the megohmmeter. For other specimens, the overall outermost shield (if any) or if there is no overall shield, the individual shields that are outermost in the specimen core shall be electrically connected to one terminal of the megohmmeter. The other megohmmeter terminal shall be electrically connected to the water surrounding the specimen (as by means of a bare metal wire which is both connected to

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the megohmmeter terminal and suspended within the water surrounding the specimen). The resistance measurement shall then be performed as specified. The temperature of the water surrounding the specimen shall also be measured, and shall be used to determine F in the following formula:"

PAGE 88

* 4.8.11.2, first sentence in equation: Delete "= 185" and substitute "= -185".

PAGE 90

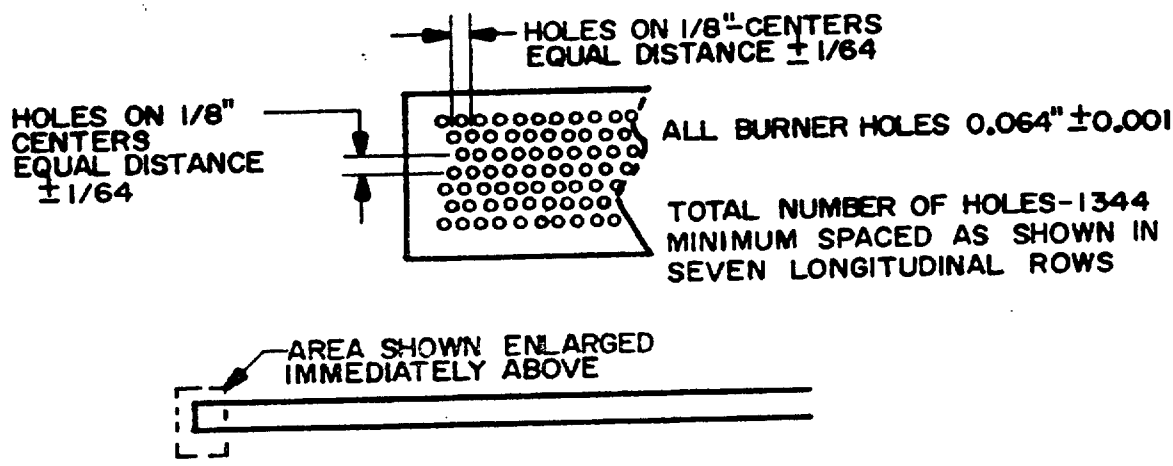
* 5.1.2, first sentence: change "XXII" to "XXIII".

TABLE XXII, Change to "TABLE XXIII".

PAGE 99

* FIGURE 8: Delete and substitute:

"



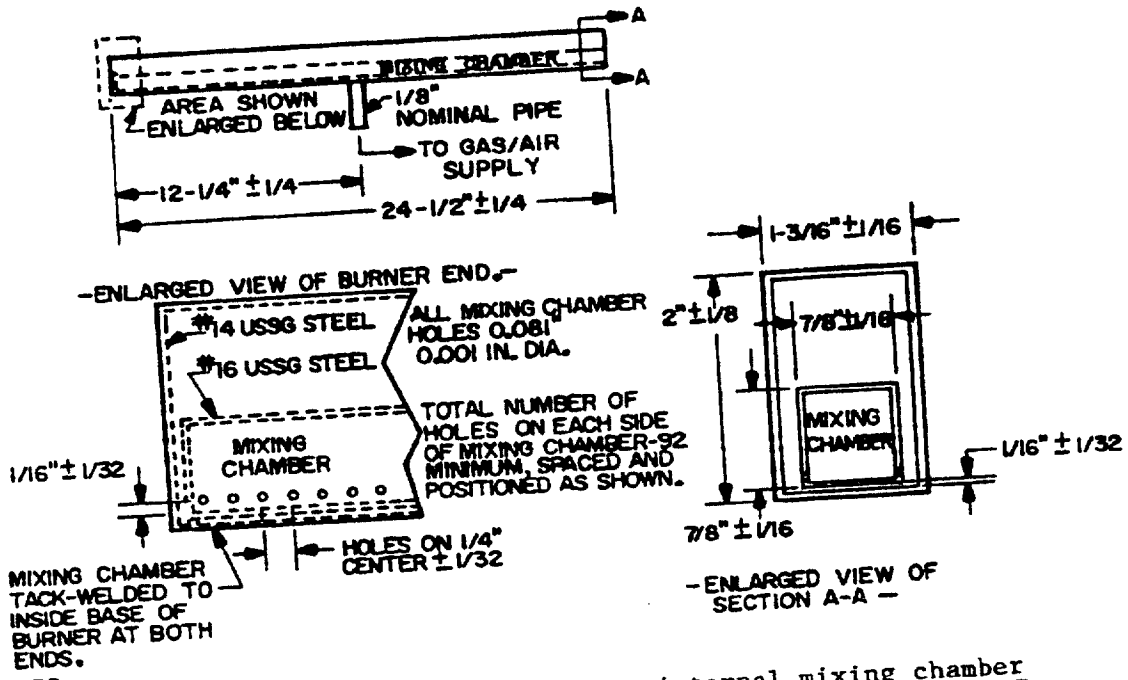
SH 13203076

FIGURE 8. Top view of burner for the gas flame test."

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* FIGURE 9: Delete and substitute:

"



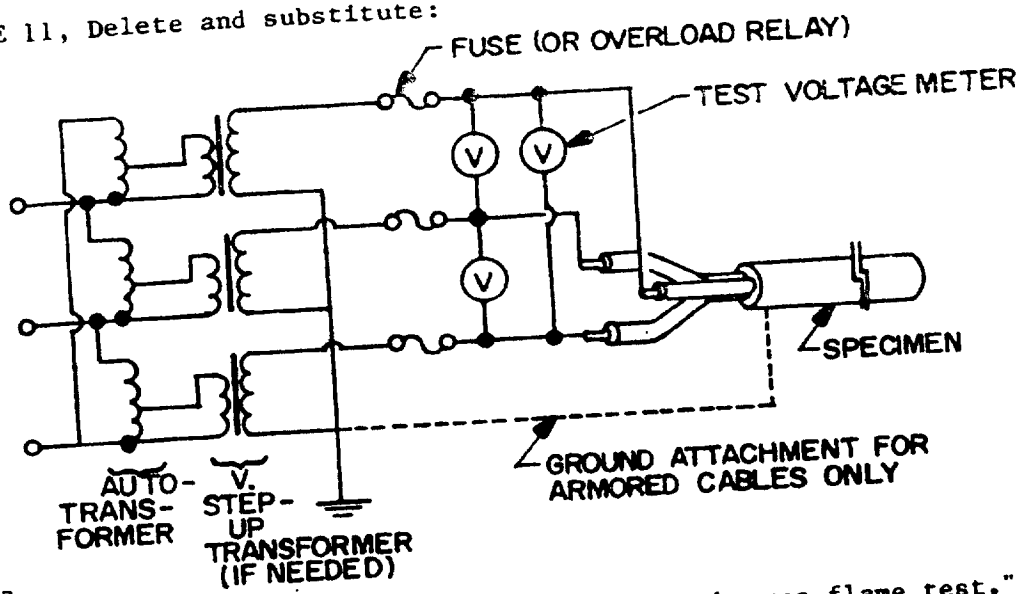
SH 13203077

FIGURE 9. Side view of burner showing internal mixing chamber for the gas flame test."

* FIGURE 10, Delete.

* FIGURE 11, Delete and substitute:

"



SH 12747

FIGURE 11. Application of test voltages for the gas flame test."

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